D02 · Planning Maturity & Digital Transformation

04 · AI / ML in planning — what works, what does not

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1 WHAT IT IS

A disciplined AI/ML-in-planning practice asks not "should we use ML" but "where does ML beat the baseline" — selecting techniques by where they demonstrably outperform established statistical methods, benchmarking every model against a simple baseline, and deploying only where the lift is real and the data supports it, instead of applying ML by hype to problems a simpler method already solves better.

2 WHY IT MATTERS

Makridakis, Spiliotis & Assimakopoulos 2018 **(Makridakis et al., 2018 — peer-reviewed)**, comparing machine-learning and statistical forecasting methods, found that the ML methods did not systematically outperform far simpler statistical ones and were often beaten by them — while costing far more in computation and complexity. The result is not anti-ML; it is that the technique must be matched to the problem, and complexity is not a proxy for accuracy. Planning organizations routinely ignore this, applying black-box ML to smooth series a simple model forecasts better, or launching ML on data that cannot support it. The pilots fail to beat the baseline they were never measured against, and each failure erodes leadership's trust — so that **the genuine ML opportunities, where the lift is real, get starved of credibility by the hype-driven ones that never should have launched**.

3 HOW IT IS DONE

1. **Start from the problem, not the technique.** Define the planning problem and ask what method fits it, rather than starting with "we want to use ML." Technique-first projects look for a problem to justify the tool; problem-first projects find the method — sometimes ML, often not — that actually wins.
2. **Benchmark every model against a simple baseline.** The naive seasonal forecast and a standard statistical model are the bar. An ML model that does not beat them out-of-sample has failed regardless of its sophistication, and deploying it anyway is paying more for less.
3. **Target ML where it has a structural edge.** ML earns its complexity on problems with many interacting drivers, non-linearities, rich external signals, or high-dimensional patterns — demand sensing, complex promotions, image/text inputs. It rarely beats a good statistical model on a clean, smooth univariate series. Match the method to the problem's shape.
4. **Verify the data can support the model.** ML is more data-hungry and more sensitive to data quality than statistical methods. On fragmented or thin data, a complex model overfits and underperforms. Confirm the integrated data foundation exists before launching the model that depends on it.
5. **Keep the model interpretable enough to trust and govern.** A black-box forecast planners cannot interrogate will be overridden or ignored, destroying its value. Favor models whose drivers can be explained, and measure forecast value-add so the model's contribution is visible and accountable.
6. **Industrialize only what survives the benchmark.** Move a model from pilot to production only after it beats the baseline on out-of-sample data with a maintainable pipeline. Scaling a model that never cleared the bar institutionalizes the cost and the distrust; scaling a proven one is where the real value compounds.

WORKED EXAMPLE — ILLUSTRATIVE

A company launched a deep-learning forecasting initiative across the entire catalog — **baseline: never benchmarked against the existing statistical forecast, beaten by it on the smooth A-items, ML pilot declared a failure, leadership trust in analytics damaged**. It restarted problem-first: kept statistical methods for smooth series, targeted ML only at demand sensing and complex promotions where it had an edge, benchmarked everything against the naive and statistical baselines. Results: **ML deployed on the ~20% of problems where it genuinely beat the baseline, measurable accuracy lift there, and analytics credibility rebuilt** — because the technique was finally matched to the problem.

IN PRACTICE

Baseline-benchmark every model

Make beating the naive and statistical baseline out-of-sample a mandatory gate before any model advances. The benchmark is the single discipline that prevents complexity from masquerading as value; no model earns deployment without clearing it.

Match technique to problem shape

Reserve ML for problems with the structure that rewards it and keep statistical methods where they win. A portfolio that uses the right method per problem outperforms one that applies a single fashionable technique everywhere — and costs less to run and maintain.

Measure forecast value-add on every model

Track whether each model — and each human override of it — adds or destroys accuracy versus the baseline. FVA makes the model's real contribution visible and turns "we use AI" into "the AI adds X points of accuracy here," which is the only claim worth making.

Sequence ML after data and process maturity

Deploy ML only once the integrated data layer and sound process exist beneath it. ML on fragmented data and a broken process amplifies the dysfunction and fails the benchmark; the maturity sequence — process, data, then advanced analytics — is what gives ML the foundation it needs to actually win.

WHAT YOU WOULD USE

AI / ML IN PLANNING

o9 / SAP IBP / Kinaxis — embedded ML in planning — Planning platforms with embedded ML for demand sensing, forecasting, and pattern detection, governed within the planning workflow and benchmarked against statistical baselines. **(o9, SAP, Kinaxis — vendor)** The choice when ML should live inside the plan with FVA and benchmark discipline built in.

Blue Yonder / RELEX — applied ML for retail/CPG — Strong applied ML for demand sensing, promotions, and replenishment where the problem structure rewards it. **(Blue Yonder, RELEX — vendor)** A solid fit where ML targets the high-driver problems — promotions, store-level sensing — it genuinely improves.

Data-science platforms (Databricks, open-source ML) — General ML and data-science platforms for custom models where packaged planning ML falls short. **(Databricks, open-source frameworks — vendor)** The vertical option when the ML problem is bespoke and the team has the capability to build and govern it rigorously.

For operations that don't need ML yet — Well-chosen statistical methods (Python statsforecast, R fable) benchmarked rigorously — which beat misapplied ML on most planning problems. The honest answer for many organizations is that the baseline is the right tool; ML is the next step only where a benchmarked pilot proves it wins.

4 SO WHAT

Operations that benchmark ML against a simple baseline and deploy it only where it wins capture real lift and protect their analytics credibility; operations that apply ML by hype launch pilots that lose to a naive forecast and burn the trust the genuine opportunities needed. The most expensive AI mistake in planning is not under-investing — it is deploying complexity where a simple method already wins, then mistaking the failure for proof that AI does not work. Match the technique to the problem, measure against the baseline, and let the lift, not the hype, decide.

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